CHAPTER 5:

# Big Rip

> I keep thinking about this river somewhere, with the water moving really fast. And these two people in the water, trying to hold onto each other, holding on as hard as they can, but in the end it's just too much. The current's too strong. They've got to let go, drift apart. That's how it is with us.
>
> Kazuo Ishiguro, *Never Let Me Go*

For a cosmic phenomenon that is arguably the most important thing in the universe, dark energy is surprisingly difficult to study. As far as we can tell, it exists everywhere in the universe, completely uniformly, woven into the fabric of space itself, and its only effect is to stretch space out so gradually that it has no detectable impact on any scale smaller than the vast expanses between distant galaxies. Dark matter physicists have it much easier—despite being just as invisible as dark energy, dark matter makes its presence very known by clumping around virtually every galaxy or cluster of galaxies we've ever seen, dominating the gravitational field, bending light, and altering the course of cosmic history from the very beginning. Dark energy, on the other hand, just … expands.

This doesn't completely prevent us from studying it. There are essentially two handles we have on dark energy: the expansion history of the universe and the way that galaxies and clusters of galaxies have grown over time. For both of these, we're peering into the distance and the past, tracing out the evolution of the cosmos over time. But no matter how we look, we're trying to tease out small effects using faint signals and statistics.

As challenging as these kinds of studies are, it's worth putting in the effort, since dark energy is both the dominant component of the cosmos and a sure sign of some new physics beyond our current understanding.

That, and the fact that depending on what dark energy turns out to be, it might violently and inescapably destroy the universe, much sooner than anyone ever imagined. Why wait for the slow fade of a Heat Death, if you can have a dark energy apocalypse as sudden and dramatic as the